

CS1540 – Midterm Model Solutions (with Grading Notes)

1. Constructing Broadcast Towers

General Grading Remarks

- The intended model assumes the highway is **continuous** and towers **may overlap**.
- If a solution instead treats the highway as **discrete** (e.g., positions $0, 1, 2, \dots$), this is acceptable with penalty for question a. and acceptable for the rest provided the next points.
- In a discrete interpretation, the greedy step may use $p + 1$ instead of p when extending coverage.
- Towers are intended to **overlap**. If a solution assumes **non-overlapping towers**, the greedy strategy is generally incorrect. This version of the problem requires dynamic programming.
- In the given example, continuous non-overlapping coverage is impossible, but a discrete non-overlapping solution such as $\{[0, 3], [4, 10]\}$ may appear.

(a) [2 points]

An optimal solution is:

$$\{[0, 5], [4, 10]\}.$$

These intervals cover $[0, 10]$ and use the minimum number of towers.

Grading note:

- If a solution gives a discrete non-overlapping solution (e.g., $[0, 3], [4, 10]$), a penalty of -1 is applied, unless eventual explanations go against instructions of the exercise, in which case 0 points are given.

(b) [7 points] Greedy Rule

Partial Solution [1 points]

A partial solution is a set of towers covering a prefix $[0, p]$ of the highway.

Greedy Rule [6 points]

At each step, choose among all towers with $a_i \leq p \leq b_i$ the one with maximum b_i . Update $p := b_i$.

Discrete variant (acceptable, but I believe no one did it):

If using discrete positions, define coverage up to p , and choose a tower covering $p + 1$ with maximum b_i .

Grading note:

- Accept either continuous or discrete formulation if consistent.
- Points were deducted if the correct idea was conveyed but with erroneous or ambiguous formalism. Common mistake: Given two intervals I and J saying $I < J$ makes no sense...

(c) [8 points] Correctness Proof

Setup [2 points]

Let the greedy algorithm choose towers

$$G_1, G_2, \dots, G_k.$$

Let $OPT = (O_1, O_2, \dots, O_m)$ be an optimal solution using the minimum possible number of towers.

Among all optimal solutions, **CHOOSE ONE THAT AGREES WITH THE GREEDY SOLUTION FOR AS MANY INITIAL CHOICES AS POSSIBLE**. That is, choose OPT so that for some maximum index $r \geq 0$,

$$O_1 = G_1, O_2 = G_2, \dots, O_r = G_r.$$

We will show that in fact OPT must agree with Greedy everywhere, and hence the greedy solution is optimal.

Grading note:

- The part in upper case is the essential part of your demonstration, without it the setup points were not given.

Exchange argument [6 points]

Assume for contradiction that OPT does not fully agree with Greedy, so $r < k$ and

$$O_{r+1} \neq G_{r+1}.$$

Let p be the furthest point covered by the first r towers:

$$[0, p] \text{ is covered by } G_1, \dots, G_r$$

(and also by $O_1, \dots, O_r$, since these towers are the same).

Now consider the next greedy choice G_{r+1} . By definition of the greedy rule, G_{r+1} is a tower containing p whose right endpoint is as large as possible.

Since OPT is a feasible solution and its first r towers cover only up to p , its next tower O_{r+1} must also contain p ; otherwise the highway would not remain covered at point p .

Therefore both G_{r+1} and O_{r+1} are candidates for the next step, and by the greedy choice,

$$b(G_{r+1}) \geq b(O_{r+1}).$$

Now replace O_{r+1} in OPT by G_{r+1} . The resulting set of towers still covers the highway:

- the first r towers still cover $[0, p]$;
- G_{r+1} covers at least as far to the right as O_{r+1} ;
- so all later towers in OPT can still cover whatever remained covered before.

Thus we obtain another optimal solution with the same number of towers as OPT , but which agrees with the greedy solution for the first $r + 1$ choices. This contradicts the choice of OPT as agreeing with Greedy for as many initial choices as possible.

Hence no such disagreement can occur. Therefore OPT agrees with Greedy at every step, and the greedy algorithm is optimal.

(d) [3 points] Runtime

Finding the next tower requires scanning all n towers:

$$O(n).$$

Grading note:

- Accept $O(n \log(n))$ with clear statement that the selection of the next tower is at most $O(n)$.

2. Recurrences and Master Theorem

(a) [4 points] Algorithm A

$$T(n) = 7T(n/4) + \Theta(n). \quad [2 \text{ points}]$$

$$T(n) = \Theta(n^{\log_4 7}). \quad [2 \text{ points}]$$

(b) [4 points] Algorithm B

$$T(n) = 5T(n/5) + \Theta(n \log n). \quad [2 \text{ points}]$$

$$n^{\log_5 5} = n.$$

$$T(n) = \Theta(n \log^2 n). \quad [2 \text{ points}]$$

Grading note:

- $T(n) = \Theta(n \log n)$. resulted in a deduction of 1 point.

(c) [2 points] Best Algorithm

$$\Theta(n \log^2 n) \ll \Theta(n^{1.404}) \ll \Theta(n^{1.585}).$$

Thus:

Algorithm B